Journal of Investigative Surgery

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/iivs20

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To cite this article: Burak Özkan, Atilla Adnan Eyüboğlu, Aysen Terzi, Eda Özturan Özer, Burak Ergün Tatar & Cagri A. Uysal (2022): The Effect of Adipose Derived Stromal Vascular Fraction on Flap Viability in Experimental Diabetes Mellitus and Chronic Renal Disease, Journal of Investigative Surgery, DOI: 10.1080/08941939.2022.2066741

To link to this article: https://doi.org/10.1080/08941939.2022.2066741

Published online: 21 Apr 2022.

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JOURNAL OF INVESTIGATIVE SURGERY
https://doi.org/10.1080/08941939.2022.2066741

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ORIGINAL RESEARCH

The Effect of Adipose Derived Stromal Vascular Fraction on Flap Viability in Experimental Diabetes Mellitus and Chronic Renal Disease

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ABSTRACT

Background: The presence of chronic renal disease(CRD) concurrently with diabetes mellitus(DM) increases the flap failure. Adipose derived stromal vascular fraction (SVF) is known to enhance skin flap viability in both healthy and diabetic individuals. The aim of this experimental study was to investigate the effect of SVF on skin flap viability in rats with DM and CRD.

Methods: 48 Sprague-Dawley rats were separated into four groups as follows: group I (control), group II (diabetes mellitus), group III (chronic renal disease), and group IV (diabetes with chronic renal disease).Two dorsal flaps were elevated. Flaps on left side of all groups received 0.5 cc of SVF, while same amount of plasma-buffered saline (PBS) was injected into right side. On postoperative day 7, flaps were harvested for macroscopic, histopathologic and biochemical assessments. Areas of flap survival were measured macroscopically. Blood level of vascular endothelial growth factor (VEGF) was measured after injection of SVF.

Results: Macroscopically, SVF has significantly improved flap viability ($p < 0.05$). Flap viability percentage was lower in DM and CRD groups when compared with healthy control group. In respect of new capillary formation, there was a statistically significant difference between SVF injected flaps and PBS injected sides ($p < 0.05$). Similarly, VEGF levels were higher in all study groups and there was a significant difference in comparison to control group ($p < 0.05$).

Conclusions: The study showed that injection of SVF increased flap viability via endothelial differentiation and neovascularization. In vivo function of stem cells might be impaired due to uremia and diabetes-related microenviromental changes.

ARTICLE HISTORY
Received 28 February 2022
Accepted 11 April 2022

KEYWORDS
Flap viability; diabetes mellitus; renal disease; stromal vascular fraction; stem cell; microsurgery

Introduction

Diabetes mellitus (DM) is still considered one of the leading causes of foot ulcers in the world. Almost one quarter of diabetic patients develop foot ulcers during their life time [1]. The concurrence of diabetes mellitus and chronic renal diseases (CRD) increases the risk of development of diabetic foot compared to diabetic patients with normal renal function. The risk of lower limb amputation in patients with renal failure is 10 times greater than that of diabetic patients without uremia [2]. Reconstruction of chronic wounds in patients with diabetic nephropathy has been a challenge for surgeons due to impaired wound healing capacity and the complexity of the wound itself. Treatment modalities include skin graft, local skin flaps, local or distant muscle or skin flaps. Skin flap surgery is indicated to cover exposed bones or tendons, or to cover deep tissue defects on weight bearing areas under the foot where skin grafts might be thin to withstand the applied pressure by the weight. Success rates in flap surgery are low compared to normal population [3]. Several studies have discussed different strategies to enhance skin flap circulation in diabetic individuals, such as atorvastatin, all-trans retinoic acid and adipose derived stem cells (ADSCs) [4–6].

Adipose tissue is rich in progenitor cells [7]. Zuck et al. have demonstrated stem cells in adipose tissues can be processed effectively and easily [8]. On the other hand, stromal vascular fraction (SVF), which is derived from adipose tissue following several processes, has a high concentration of stem cells, growth factors, macrophages and platelets. SVF has tissue regeneration capacity to repair damaged tissues, beside its effect on endothelial proliferation and neovascularization [9]. This property is being increasingly investigated in the field of skin flap survival. Current literature presents studies

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This article was presented in EURAP S Research Council 2016, Brussels, Belgium and P RSC 2017 Durham, USA.
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on the effect of stem cells on skin flap viability in diabetic models. It has been demonstrated that local transplantation of autologous adipose-derived stem cells is increasing the viability of random pattern skin flaps through both mechanisms of vasculogenesis and angiogenesis [10]. However, effects of stromal vascular fraction or cultured mesenchymal cells on skin flap viability in the presence of diabetes mellitus and chronic renal disease has not been studied yet [6, 11, 12].

In this study, we aimed to investigate the effects of locally-injected adipose derived stromal vascular fraction on skin flap viability in rat models of diabetes mellitus and CRD.

Material and method

Ethical approval

This experimental study was approved by the Local Ethics Committee for Animal Experiments at our university. (project number DA14/17 – 21/10/2014) All surgical procedures were performed at the same center according to the guidelines for the care of laboratory animals published by the United States National Institutes of Health.

Animal model

The study included a total of 48 male Sprague-Dawley rats with a weight of 400-450 grams The rats were kept in 23 °C with 12 hours of light and dark cycles. Rats consumed standard raw chow and water *ad libitum*. Postoperative analgesia and antibiotics were administered.

Sample size estimation

The mean numbers of size from the previous studies evaluating flap viability in terms of blood VEGF levels,vascular density,flap viability and immunochemical were set into parameters in the G * Power (v3.1.9.2) program [13–15]. The desired α error was set at 0.05, β error 80%. As a result of the calculation, we found that 48 rats would provide 90% statistical power at a 5% significance level.

Experimental protocol

Rats were randomly divided into four groups, with twelve rats in each group.

- In group I(control), The skin flaps were elevated. Phosphate buffered saline (PBS) were injected to both flaps.
- In group II(DM), Diabetes mellitus model created with streptozotocin, after 3 weeks two rats were used for stem cell harvesting procedure.The skin flaps were elevated.SVF and PBS were injected to flaps.
- In group III(CRD),Chronic Renal Dissease(CRD) was created with 5/6 nephrectomy,after 6 weeks two rats were used for stem cell harvesting procedure.The skin flaps were elevated.SVF and PBS were injected to flaps.
- In group IV(DM + CRD), DM and CRD models were created like group II and group III, respectively and after 6 weeks two rats were used for stem cell harvesting procedure.The skin flaps were elevated.SVF and PBS were injected to flaps.
- Weights of the animals were measured in the beginning and at the end of the experiment.
- For group III and group IV, the blood urea, creatinine and hemoglobin values were measured.
- Blood VEGF values were measured on the 1st and 7th day of flap elevation.
- On day 7 of flap elevation, flap viability was measured.
- The animals were sacrificed on the 7th postoperative day,and specimens were examined histologically and immunochemically.

Induction of diabetes mellitus

Intraperitoneal instillation of 65 mg/kg Streptozotocin (Sigma Chemical Co. St. Louis Missouri, USA) dissolved in isotonic saline solution was performed for diabetes induction. Forty-eight hours later, 1 cc of blood was taken from the tail vein and plasma glucose levels were measured using blood glucometer (Wellion, Austria). Animals with blood glucose levels above 200 mg/dl were accepted as diabetics. Diabetic animals were followed for 3 weeks before the intervention [14].

Induction of chronic renal disease

Chronic renal disease was induced through 5/6 nephrectomy (unilateral nephrectomy and removal of two thirds of contralateral kidney) under anesthesia provided by 40 mg/kg ketamine hydrochloride (Pfizer Inc.) and 5 mg/kg xylazine hydrochloride (Rompun, Bayern Inc.) [16]. (Figure 1) A total of 300 µl of blood was taken. Blood hemoglobin (TEST-1, Alifax), creatinine and urea nitrogen levels of tail vein (Architect c8000, Abbott Diagnostics) were assessed at weeks 0, 3 and 6 to confirm the development of renal impairment. Animals were followed for 6 weeks following subtotal nephrectomy.

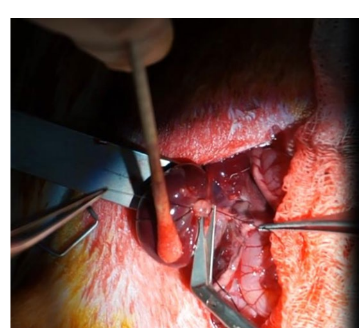

Figure 1. Chronic renal disease was induced through 5/6 nephrectomy.

Isolation and preparation of stromal vascular fraction(SVF) cells

Two rats of each group(II-III-IV) were used to obtain SVF. Anesthesia was carried out by injection of 40 mg/kg ketamine hydrochloride (Ketosal; Interhas Co. Ltd., Ankara, Turkey) and 5 mg/kg xylazine hydrochloride (Xylazine Bio; Interhas Co. Ltd., Ankara, Turkey) via intraperitoneal injection. Both inguinal regions were shaved and cleaned using 10% povidone-iodine. Stromal vascular fraction cells were harvested and processed according to previous studies [17, 18]. Briefly, the inguinal fat pads of the rats were excised and washed extensively with phosphate-buffered saline (PBS) (Gibco BRL, Grand Island, NY). Fat pads were minced finely on 100 mm^2 tissue culture plates (Marienfeld, Konigshofen, Germany) and the tissue was then rinsed three times in phosphate-buffered saline for 5 minutes followed by enzymatic digestion with 0.15% collagenase (Serva, Heidelberg, Germany) and vigorous shaking(with vortex) for 30 minutes at 37 °C in a 50 cc centrifuge tube. Next, an equal volume of control medium (Dulbecco's Modified Eagle Medium (DMEM)) (Gibco BRL), 10% fetal bovine serum (FBS) (Gibco BRL), and 1% antibiotic-antimycotic solution (Bilim Ilac, Istanbul, Turkey) was added to neutralize the collagenase.

Then, the cell suspension was centrifuged at 1300 rpm (260 g) for 5 minutes and the cell pellet was re-suspended with a control medium. After cell counting using methylene blue, the cells were ready for 1,10-Dioctadecyl–3,3,30,30-tetramethylindocarbocyanine (DiI) labeling.

1,1′-Dioctadecyl–3,3,3′,3′-tetramethylindocarbocyanine labeling of stromal vascular fraction cells

1,10-Dioctadecyl–3,3,30,30-tetramethylindocarbocyanine (DiI) (Invitrogen; Medsantek, Life Technologies, Carlsbad, CA) was dissolved in 99% ethanol at a concentration of 25% and stored at −20° C for use. Cells were labeled with fluorescent DiI for paraffin slides according to the manufacturer's recommendations. Cells in suspension were incubated with DiI at a concentration of 2.5 lg/ml in phosphate-buffered saline for 5 minutes at 4° C, as described previously by Uysal et al. [19]

Skin flap elevation

A modification of Ichioka's dorsal island flap was used as the flap model [20]. Two caudally based, 1 × 5 cm sized dorsal random skin flaps were elevated including panniculus carnosus layer (Figure 2). 0.5 cc of SVF (5 x10^7cells) was injected subcutaneously to five different areas of left flap from proximal to distal while same amount of PBS was injected to the right side of the same animal (Figure 3). Flap donor sites were closed primarily. Both flaps were sutured to the dorsal skin to prevent diffusional nourishment from the base (Figure 4).

Figure 2. Two caudally based, 1x5 cm sized dorsal random skin flaps were elevated.

Figure 3. SVF and PBS were injected to flap.

Figure 4. The flaps were sutured.

Measurement of blood VEGF levels

Blood VEGF levels of all groups were measured on the day of flap elevation(just before the flap elevation) and on the post-operative seventh day. VEGF tests were conducted by ELISA cold sandwich method and VEGF Immunoassay kit (BIOSOURCE, California, USA).

Assessment of the flap viability

Areas of flap survival and necrosis were assessed on the seventh day of flap elevation and SVF implantation. Photographs of the flaps were taken from 50 cm distance (Sony-α Nex-6 SLR, 3.5- 5.6/16-50 lens) and they were

Figure 5. Areas of flap survival and necrosis were assessed with software program. (Adobe Photoshop 7.0).

evaluated with software program (Adobe Photoshop 7.0) (Figure 5). Pixels of the viable and non-viable areas were calculated in the measurement log of the program and flap survival percentage was measured, as described previously [17]. Finally,the rats were sacrificed with high dose ketamine. Tissue specimens were taken for analysis.

Assessment of vascular density

Tissue specimens were harvested as longitudinal slices from each flap on postoperative 7th day. Tissue specimens were fixed in 10% neutral buffered formalin for 24 hours. Tissue materials were embedded in paraffin and sectioned to 4-6 µm width. Microscopic stains included Hematoxylin & eosin (H&S) for vascularity. The stained materials were reviewed by pathologists under light microscope (Olympus BX51, Tokyo, Japan). New capillary formations were counted and noted in 20 different sections of the slide (40x magnification), as described by Eyuboğlu et at [19].

Glomerular analysis

The remaining 1/6 kidneys of animals in Group III and Group IV were excised after sacrification for histopathological documentation and demonstration of chorinic renal disease(CRD) development in comparison to kidney specimens taken from animals of healthy control group. Tissue materials were sectioned to 4-6 µm width and stained with H&E. The glomerular tuft was measured in 20 areas in Adobe Photoshop 7.0 software program and evaluated for changes such as mesangial expansion and glomerular hypertrophy.

Immunohistochemical assessment

Tissue section were embedded in paraffin and immunohistochemical staining was done for detection of monoclonal Cluster of Differentation(CD)31 antibodies (1:100, DAKO, Denmark) using streptavidin-biotin-immunoperoxidase method. Slices were dewaxed in xwlene and rehydrated in graded alcohol (96%, 90%, 80%, 70%). Then, 3% hydrogen peroxide was applied for 5 minutes for blocking of endogenous peroxidase activities. Slices were then immersed in a thermostatic bath containing preheated ethylene diamine tetra acetic acid (EDTA) for 15 mins and then left for cooling down at room temperature. Primary CD31 antibody was added and incubated for 2 hours at room temperature.

Monitorization of 1,1′-Dioctadecyl-3,3,3′3′-Tetramethylindocarbocyanine perchlorate (DiI) labelled cells

Tissue specimens were stained with hematoxylin only. The remaining slides were labeled with hematoxylin and eosin together. Capillaries counted in the hematoxylin-only slides that were adjusted by the findings in other slides. The capillaries were then assessed under immunofluorescent microscopy (Nikon Eclipse E600) at light spectrum of 565 nm. DiI labeled endothelium was examined under 20x, 40x, 60x, and 100x magnifications, as prescribed by Uysal et al. [17] Digital images were reviewed with Cytovision Genus Software program.

Statistical analysis

The data was evaluated in SPSS program (SPSS Ver. 17.0, SSPS Inc, Chicago IL, USA) Shapiro-Wilk test was used to check the consistency of the variables. The homogeneity of the group variance was evaluated with Levene Test. Then, the three groups of variables were analyzed by One Way ANOVA if the prerequisites of variance analysis were met. Kruskal-Wallis Test was used if the prerequisites of variance analysis were not met. In the follow-up, Bonferroni Dunn test, one of the multiple comparison tests, was performed to determine which groups differed. When evaluating between days and groups, one of the two-factor factors, Repeated Measure Analysis (Repeated Measures) was used. As a result of Mauchly's sphericity test, which is a prerequisite for repeated measurement variance analysis, it was determined that the Group * Day interaction sphericity assumption was met. Therefore, it was determined that the results obtained from the changes of the groups according to the days in the blood VEGF levels were sufficient for the comparison of the groups, in addition, it was not necessary to compare the values between the groups within the same day. Percentage of flap survival, capillary density, glomerular analysis was evaluated with One Way ANOVA. Student's t-test was used for weight changes and kidney function testes. P values less than 0.05 were considered as statistically significant.

Results

No rats died during the study. No flap complications such as infection, hematoma and seroma were encountered.

Weight change results in rats

Weights of the animals were measured in the beginning and at the end of the experiment. Weight loss in groups II, III and IV was statistically significant (Table 1).

Measurement of blood parameters

The increase in blood urea, creatinine and decrease in hemoglobin in groups III and IV at the 3rd and 6th weeks were statistically significant when compared with the beginning of the experiment ($p < 0.05$) (Table 2).

Assesment of blood VEGF levels

The increase in VEGF levels between days 0-7 was statistically significant in group II only.($p < 0.05$) There was no significant difference between blood VEGF levels between day 0 and day 7 in both group III and IV ($p > 0.05$) (Table 3).

Flap viability measurement

The flap viability percentages of the groups are summarized in Table 4. In Group I(control), PBS was applied to both flaps and there was no significant difference in flap viability. ($p > 0.05$) A significant difference was detected between the PBS applied flaps in Group II-III-IV and the PBS applied flap in Group I.($p < 0.05$) A statistically significant difference was found between the flaps in which PBS was applied and the flaps in which SVF was applied in group II-III-IV. ($p < 0.05$)

Histological and immunohistochemical findings

Capillaries were counted using H&E and CD31 antibody staining. Capillary densities of the groups are summarized in Table 5. Vascular density in the SVF flaps was found to

Table 1. Comparison of weight changes(gr) of rats in groups.

	Before Experiment (Mean + SD(Max-Min))	After Experiment(Mean + SD(Max-Min))	p value
Group I(Control)	432.3 ± 21(456-400)	426.2 ± 22(450-385)	>0.05
Group II(DM)	420.8 ± 75(452-380)	347.2 ± 28(386-302)	>0.05
Group III(CRD)	436 ± 16(454-404)	280.4 ± 26(314-241)	**<0.05***
Group IV(DM + CRD)	413.6 ± 20(452-386)	277.3 ± 22(304-238)	**<0.05***

*:Statistically significant.
-The parameters were presented as mean ± standard deviation(SD).
-DM:Diabetes Mellitus
- CRD:Chronic Renal Dissease

Table 2. Comparison of blood parameter values of group III and group IV.

	Blood Urea(mg/dl)			Hemoglobin(g/dl)			Creatinine(mg/dl)			p value
	Beginning	3rd week	6th week	Beginning	3rd week	6th week	Beginning	3rd week	6th week	
Group III	17.6 ± 1.3	51.8 ± 20	71.6 ± 13.5	16.8 ± 0.8	12.75 ± 0.8	11.7 ± 1.2	0.36 ± 0.05	0.8 ± 0.1	1.02 ± 0.1	>0.05**
Group IV	15.5 ± 2.3	54.6 ± 17.9	77.6 ± 13.8	17.5 ± 0.6	14.1 ± 1.2	12.5 ± 1.05	0.4 ± 0.03	0.8 ± 0.2	1.2 ± 0.2	<0.05*

-The parameters were presented as mean ± standard deviation(SD).
-*:Statistically significant. (The difference between the beginning value of all three parameters and the 3rd/6th weeks values were significant.).
-**: Statistically not significant. (The difference between the 3rd and 6th week values of all three parameters were not significant.).

Table 3. Comparison of the changes in VEGF values in the groups.

	Blood VEGF Level(pg/ml)			
	Before Streptozotocin and nephrectomy	Just before flap elevation	7th day of flap elevation	p value
Group I	–	19.6 ± 11	12.1 ± 8	>0.05**
Group II	19.2 ± 0.45	21.9 ± 8	36.3 ± 14	**<0.05***
Group III	17.2 ± 0.337	26.8 ± 9.8	28.5 ± 6.9	
Group IV	20.2 ± 0.52	41.4 ± 36	44.1 ± 7.6	

-The parameters were presented as mean ± standard deviation(SD).
-*:Statistically significant. (The increase in VEGF levels between days 0-7 was statistically significant in group II).
-**: Statistically not significant. (There was no significant difference between blood VEGF levels between day 0 and day 7 in both group III and IV).

Table 4. Comparison of flap viability percentages in groups.

	SVF/PBS#	PBS	p value
Group I(Control)	69.5%±2.4	69.1%±2.3	>0.05
Group II(DM)	65%±7	50.7%±8.7	**<0.05***
Group III(CRD)	54.3%±2.8	42%±4.6	**<0.05***
Group IV(DM + CRD)	53%±5.8	44.3%±5.5	**<0.05***
p value	–	**<0.05****	–

-The parameters were presented as mean ± standard deviation(SD).
-*:Statistically significant. (A statistically significant difference was found between the flaps in which PBS was applied and the flaps in which SVF was applied in group II-III-IV).
-**:Statistically significant. (A significant difference was detected between the PBS applied flaps in Group II-III-IV and the PBS applied flap in Group I.).

Table 5. Comparison of flap capillary densities in groups.

	SVF	PBS	p value
Group I(Control)	4.5 ± 0.7		–
Group II(DM)	5.6 ± 1.1	3.3 ± 0.6	**<0.05***
Group III(CRD)	6.1 ± 1.2	4.1 ± 0.8	**<0.05***
Group IV(DM + CRD)	3.4 ± 0.5	2.4 ± 0.2	**<0.05***
p value	>0.05**		

-The parameters were presented as mean ± standard deviation(SD).
-:Statistically significant. (Capillary density in the SVF flaps was found to be significantly higher than PBS administered flaps in each group).
-**:Statistically not significant. (There was no significant difference between the capillary densities of the flaps in the SVF applied groups.).

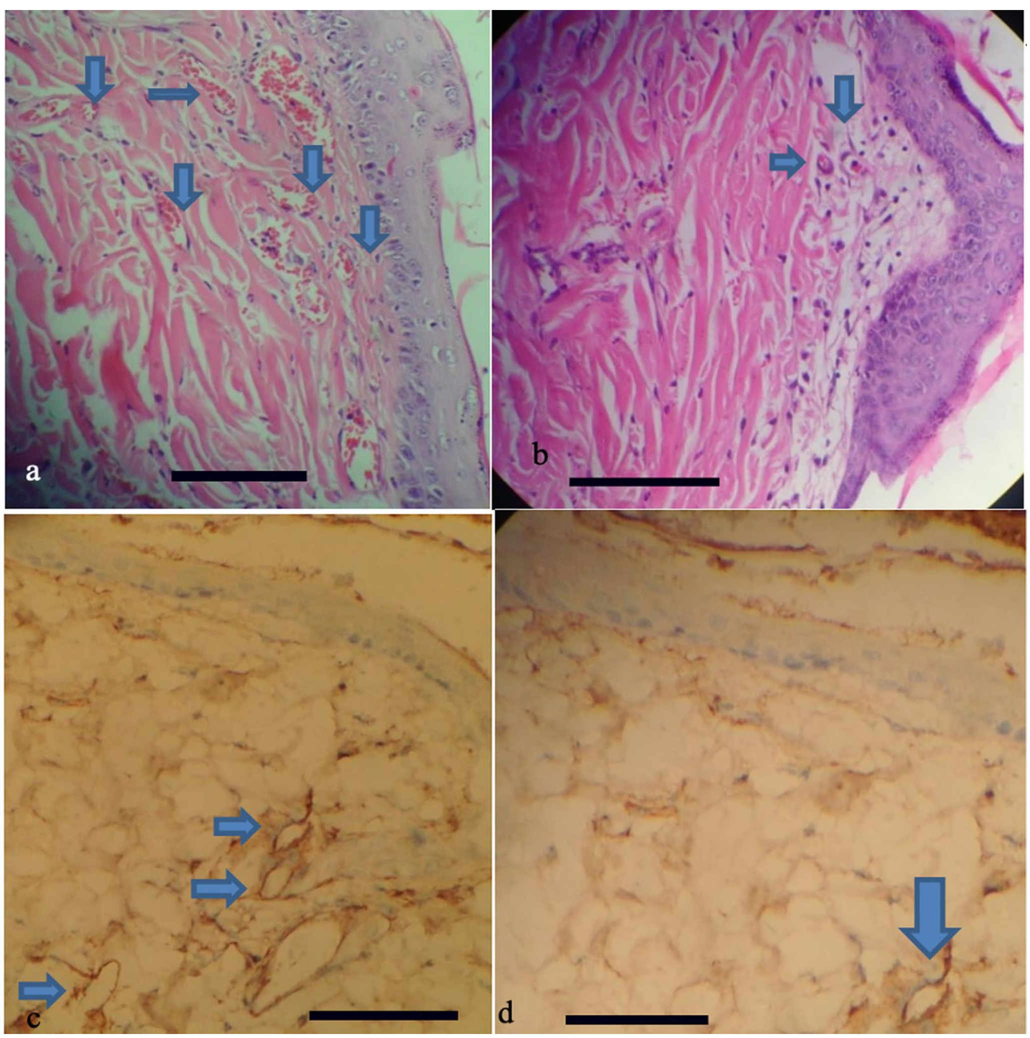

Figure 6. Histological evaluation of capillary density.**A:** Capillaries in the SVF-injected flap in group III.(Blue Arrows)**B:** Capillaries in the PBS-injected flap in group III.(Blue Arrows)**C:** CD31 staining capillaries in SVF-injected flaps in group IV. (Blue Arrows)**D:** CD31 staining capillaries in PBS-injected flaps in group IV. (Blue Arrows)

be significantly higher than PBS administered flaps in each group.(II-III-IV)(p < 0.05) There was no significant difference between the capillary densities of the flaps in the SVF applied groups (Group II-III-IV)(p > 0.05) (Figure 6)**.** The mesengium thickness in group III increased compared to group I (Figure 7)**.**

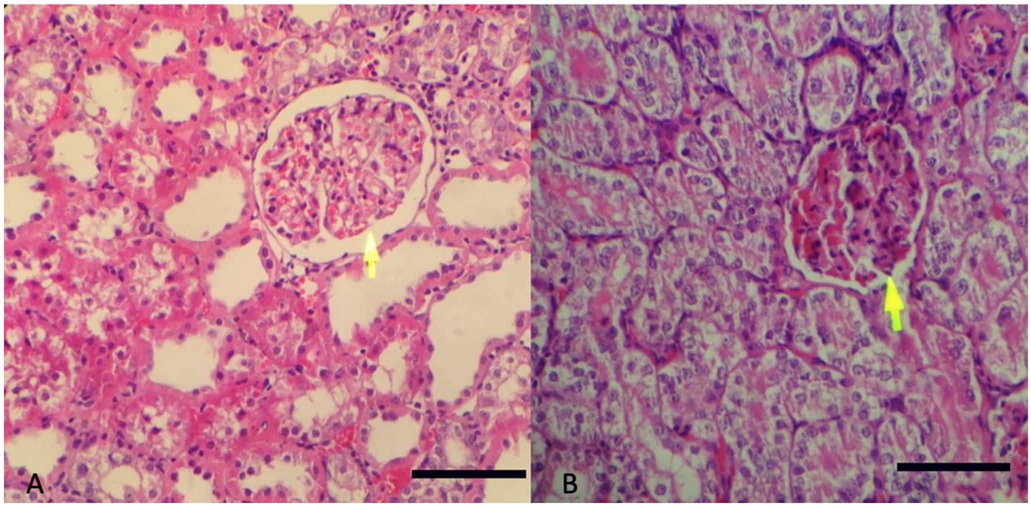

Figure 7. The mesangium thicknesses in group I and group III.**A:** Normal gloremular thickness in group I.**B:** Hypertrophy and intraglomerular fibrosis in the glomeruli in group III (Yellow arrow)

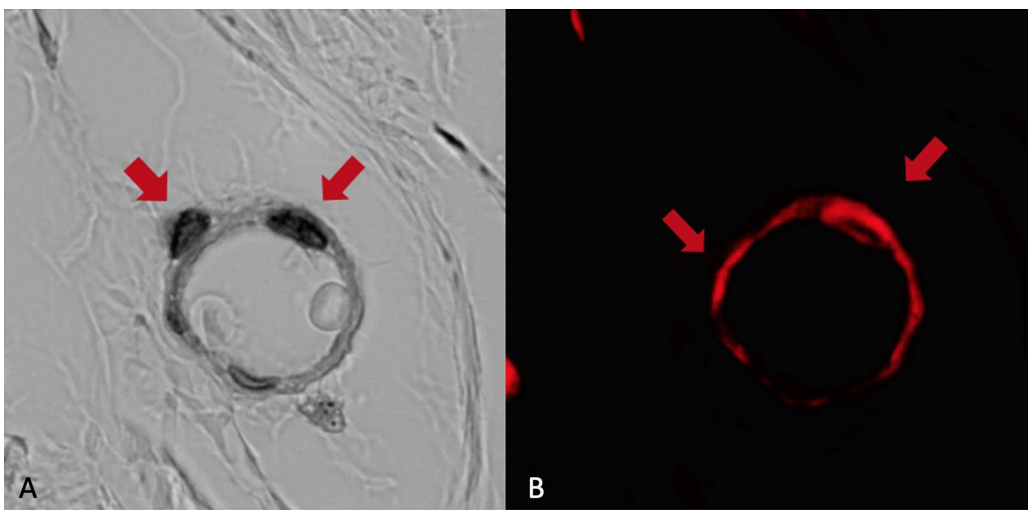

Figure 8. DiI (+) endothelial cells in group II.**A:** Light microscopy of endothelial cells(Red arrow)**B:** Immunofluorescence view of endothelial cells(Red arrow)

DiI labeled cells were found in the walls of the endothelium at 100x magnification under fluorescent microscopy (Figure 8).

Discussion

Diabetes mellitus and chronic rebal disease are systemic diseases whose incidence is increasing day by day [21, 22]. Since one of the most important causes of chronic kidney disease is diabetic nephropathy, the overlapping presentation of these two diseases is a common situation. Perfusion disorders caused by angiopathies due to these two medical conditions are mostly seen in the lower extremities. Even if the appropriate reconstruction method is used in wounds where structures such as bones and tendons are exposed, amputations may have to be performed due to the decrease in flap viability [2, 23].

Flap viability is a problem in patients with diabetes mellitus and chronic renal disease. Several theories have been proposed for the high flap necrosis rates in diabetic patients. Disruption of the neovascularization by hyperglycemia is the most accused factor in the recent years [24]. Other factors include the presence of glycosylation products, endothelial injury caused by local and systemic inflammation due to oxidative stress and products of lipid peroxidation, and micro-thrombosis caused by increased viscosity of the plasma [25]. Although repressed immune response, systemic inflammation and inclination toward infection are considered as the reasons for dysfunctional wound healing and flap necrosis in CRD, there are limited experimental studies about skin flap viability in CRD [26, 27]. Recently, stem cell studies have shown the success in improving flap viability [10, 28]. Thus, we aimed to investigate the effect of ADSCs on skin viability in animals with diabetes and chronic renal disease.

Flap viability was found lower in diabetic and CRD induced animals in our study. We showed that the administration of SVF has increased the flap viability significantly. As the flap model, the double dorsal flap defined by Ichioka was used because it allows the use of a small number of animals and allows the operation of the only variable on the flap [20]. In the control group, PBS was injected to show the effect of anatomical difference between the right and left flaps on flap viability. Another reason for not injecting left flap SVF in the control group is the publication in the literature showing the positive effect of SVF on flap viability in healthy animals [10]. Among the methods defined for iatrogenic diabetes mellitus, streptozocin, which is the most widely used, dose and mechanism of which were well described, was used. It was waited for 3 weeks, as suggested by Rendell et al., in order for the pathological changes related to diabetes mellitus to be observed and to have a significant effect on the flap [29]. For the iatrogenic chronic renal failure model, the 5/6 nephrectomy model was selected [30]. In previous studies, nephrectomy was performed in 2 sessions [31]. Seth et al. modified it as a single session, and we also used this model in our study [16]. In order not to affect the dorsal flap, the incision was made from the abdomen. Since there is no publication in the literature about flap viability with CRD, the publication about wound healing with CRD was taken as a reference and it was waited for 6 weeks [16].

Although mesenchymal stem cells are found in various tissues; bone marrow and adipose tissue are the most researched tissues [8, 32]. Stem cells harvested from adipose tissue have shown the ability of differentiation in a similar way to their bone marrow analogues [33–36]. Uysal et. al compared the effects of stem cells from bone marrow and adipose tissue, reporting that increase in capillary density was more profound in stem cells from adipose tissue origin [13]. Thus, we aimed to use adipose tissue as a source for stem cells. Clinical implication of mesenchymal stem cells have still strict restrictions [37]. Therefore, we preferred stromal vascular fraction which is a rich source of mesenchymal stem cells [19]. SVF was group specific, was harvested and transplanted to the same group in our study; with the aim of investigating the effects of SVF under hyperglycemic and uremic conditions.

In consistent with our findings, Emekli et al. showed that random flap viability was 45% in diabetic flaps [15]. Gao et al. reported that flap viability increased from 45% to 80% in dorsal skin flaps in diabetic rats treated with adipose derived stem cells [6]. However, they have harvested stem cells from healthy individuals, and they have used a 3 cm long and 1 cm wide random pattern dorsal skin flap. In contrast, a flap model with 5 cm long 1 cm wide was used and SVF was prepared from unhealthy animals in our study. We further investigated how adipose derived stem cells (ADSC) increases viability of skin flaps. It has shown that ADSC and SVF have dual effects in flap survival [17]. First is the direct effect via differentiation to different cell lines. Second is the indirect paracrine effect [19]. Thus, we searched in vivo differentiation and neovascularization parameters. DiI labeled stem cells were evaluated under immunofluorescent microscope to determine whether the vascularization was caused by direct endothelial transformation or by proangiogenic cytokines. In our study we did not count DiI labeled cells but in vivo transformation of stem cells into vascular endothelium were observed in merged images of immunofluorescent and light microscopy.

Blood VEGF level and capillary density were used for detection of neovascularization. We investigated blood VEGF levels before the experiment, on the day of flap elevation, and post-operative 7^{th} day. The change in VEGF levels between days 0 and 7 in groups III and IV was not statistically significant. A significant increase was seen in Group II. The effect of uremia on VEGF levels showed that uremia might disrupt ADSCs functions more than diabetes in vivo. The significant increase in VEGF level in diabetic rats was consistent with the increase in capillary density when compared with group III and group IV.

Capillary density was significantly higher in SVF administered flaps when compared with PBS administered flaps in all groups. This increase was in accordance with the increased flap viability in SVF administered flaps. The number of vessels observed in groups II, III and IV were lower than that in group I (control group). This might be due to negative effects of diabetes and uremia on neovascularization. Flap viability and capillary density were lowest in Group IV which might suggest that rats with diabetes and chronic renal disease have lower regenerative potential than animals with diabetes or chronic renal disease alone. According to our findings adipose derived stem cells improve flap survival even in uremic and hyperglycemic rats. The percentage of flap viability was lower than that of control group I. These results support the fact that hyperglycemia and uremic environment diminish adipose derived stem cell functions.

There are several studies investigating the effects of diabetes mellitus and chronic renal disease on mesenchymal stem cells. In consistent with our study, Gao et al. have showed that VEGF synthesis is increased in diabetic rats [6]. In contrast to our findings, hyperglycemia has been shown to decrease VEGF production in diabetic human and animal wound healing models [3] [8] .In a study to investigate ADSC in diabetic wound model, Lerman et al. reported that capillary migration of stem cells decreased under hyperglycemia [3] [9] .Tanaka et al. described that endothelial progenitor cell dysfunction might result in diminished neovascularization in diabetic animals [4] [0] .De Groot K. et al. have reported that hematopoietic and endothelial progenitor cells in the non-mesenchymal stem cells of the bone marrow showed a decreased proliferation in the setting of chronic renal disease [4] [1] .Noh et al. reported that rats with CRD for 6 weeks duration have impaired function of mesenchymal cells of bone marrow origin [4] [2] .Klinkhammer et al. showed that uremic toxins, proinflammatory molecules and reactive oxygen radicals were found to disrupt MSC differentiation and reproduction in CRD [43

].There are some limitations in our study. The study was conducted on subjects with relatively short duration of diabetes mellitus and chronic renal failure, not in rats exposed to the diseases for a long time.We only studied on random flaps. Perforator or free flaps were not included in the study.

In this experimental study, we showed that diabetes and chronic renal disease deteriorate flap viability. Although their effects are diminished by hyperglycemia and uremia, adipose derived stromal vascular fraction prepared from rats with diabetes and chronic renal disease might improve flap viability via both paracrine and direct effects.

The clinical impact of this study can be translated into clinical practice by the autologous use of stem cells injected into skin flaps immediately after flap elevation in patients with diabetic nephropathy. Putting in mind that autologous cells in these patients are less effective than healthy individuals. Further studies are needed to understand in vivo stem cell behavior under chronic hyperglycemic and uremic conditions.

As a conclusion SVF is a stem cell type with low cost, high efficiency, easy preparation and application. It can be used to increase flap viability in conditions such as Diabetes mellitus and chronic kidney disease that reduce flap viability. We believe that our study will shed light on future studies.

Acknowledgments

There is no contributor who does not meet the criteria for authorship. Hence, the section of acknowledgment is none.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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